Leveraging longitudinal diffusion MRI data to quantify differences in white matter microstructural decline in normal and abnormal aging

DB Archer[1,2], N Shashikumar[1], V Jasodanand[1], EE Moore[1], KR Pechman[1], M Bilgel[3], LL Beason-Held[3], Y An[3], A Shafer[3], SL Risacher[4,5], KG Schilling[6], BA Landman[7,8,9], AL Jefferson[1,2,10], AJ Saykin[4,5], SM Resnick[3], TJ Hohman[1,2]

[1]Vanderbilt Memory and Alzheimer's Center, Vanderbilt University Medical Center, Nashville, TN, USA; [2]Vanderbilt Genetics Institute, Vanderbilt University Medical Center, Nashville, TN, USA; [3]Laboratory of Behavioral Neuroscience, National Institute on Aging, National Institutes of Health, Baltimore, MD, USA; [4]Indiana University School of Medicine, Indianapolis, IN, USA; [5]Indiana Alzheimer's Disease Research Center, Indianapolis, IN, USA; [6]Department of Radiology, Vanderbilt University Medical Center, Nashville, TN, USA; [7]Vanderbilt University Institute of Imaging Science, Vanderbilt University Medical Center, Nashville, TN, USA; [8]Department of Biomedical Engineering, Vanderbilt University, Nashville, TN, USA; [9]Department of Electrical and Computer Engineering, Vanderbilt University, Nashville, TN, USA; [10]Department of Medicine, Vanderbilt University Medical Center, Nashville, TN, USA

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Background

Several prior studies have used diffusion MRI to investigate differences between normal and abnormal aging; however, many of these studies used conventional diffusion MRI measures and single site data. The goal of this study is to leverage multi-site harmonized diffusion MRI data in conjunction with an advanced post-processing technique [i.e., free-water (FW) correction] to quantify aging related tract-specific changes in white matter microstructure.

Participant Characteristics & Project Workflow

- Data were drawn from several well-established longitudinal cohorts of aging [Alzheimer's Disease Neuroimaging Initiative (ADNI), Baltimore Longitudinal Study of Aging (BLSA), Vanderbilt Memory & Aging Project (VMAP)]. In total, this dataset included 1,723 participants across 4,605 imaging sessions. See **Table 1** for cohort characteristics, demographics, and health characteristics.

Table 1 – Demographic and Health Characteristics

Measure	Cohort			p-value
	ADNI	BLSA	VMAP	
Cohort Characteristics				
Number of participants	706	721	296	-
Total number of sessions	1,859	1,858	888	-
Longitudinal follow-up (yrs)	1.23 (1.75)	2.42 (2.54)	2.01 (1.86)	<0.001
Demographics and Health Characteristics				
Age at baseline (yrs)	73.03 (7.72)	71.33 (10.21)	73.32 (7.26)	<0.001
Sex (% male)	51.42	44.24	57.77	<0.001
Education (yrs)	16.29 (2.62)	17.02 (2.38)	15.81 (2.68)	<0.001
Race (% non-hispanic white)	91.64	65.74	92.91	<0.001
APOE-ε4 (% positive)	42.21	26.91	35.81	<0.001
APOE-ε2 (% positive)	8.92	17.06	15.20	<0.001
Cognitive at baseline (% cognitively unimpaired)	46.74	98.34	56.76	<0.001
Converter status (% non-converter)	44.05	93.76	56.76	<0.001

- Diffusion MRI data was preprocessed using the *PreQual* pipeline and conventional microstructural maps were created [fractional anisotropy (FA_{CONV}), mean diffusivity (MD_{CONV}), radial diffusivity (RD_{CONV}), axial diffusivity (AD_{CONV})]. Conventional microstructural maps were transformed to MNI space using *ANTs*.
- FW-correction was conducted on the preprocessed diffusion MRI data to calculate 5 microstructural values [FW, FW-corrected FA (FA_{FWcorr}), FW-corrected MD (MD_{FWcorr}), FW-corrected RD (RD_{FWcorr}), FW-corrected AD (AD_{FWcorr})]. FW and FW-corrected microstructural maps were transformed using *ANTs*.
- Mean conventional and FW corrected microstructural measures were quantified in 48 white matter tractography templates, including 7 different tract groupings [association, limbic, projection, motor transcallosal (TC), occipital TC, parietal TC, prefrontal TC]. See **Figure 1A** for an illustration of all tractography templates.
- All microstructural values were harmonized using the *Longitudinal ComBat* pipeline (see example in **Figure 1B**) and linear mixed-effects regression was conducted to do the following:
  1) Evaluate main effects of aging in normal and abnormal agers
  2) Determine if there are significant interactions in aging between normal and abnormal agers
  3) Run bootstrapped analysis to determine which microstructural measures and white matter tracts are most vulnerable to abnormal aging
- Covariates included age, age^2, education, sex, race/ethnicity, *APOE*-ε4 carrier status, and *APOE*-ε2 carrier status.

Results

Finding 1: *Longitudinal ComBat* harmonization

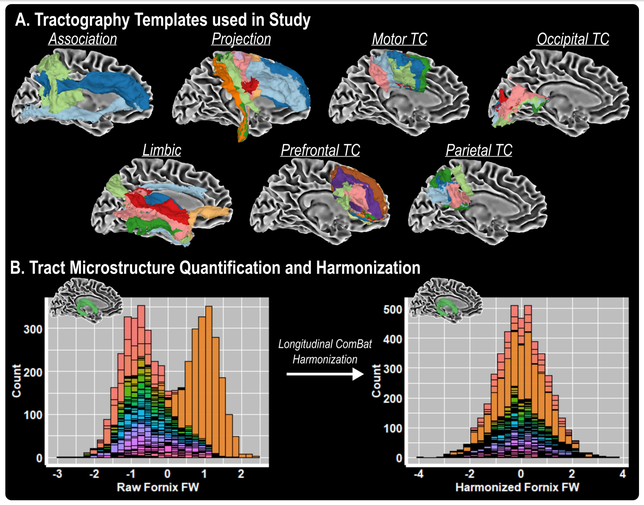

Figure 1. *Longitudinal ComBat* was used to harmonize 432 features across more than 100 site/protocol combinations and 48 white matter tracts (**A**). Harmonization for fornix FW is illustrated (**B**).

Finding 2: Strong interactions of white matter decline exist between normal and abnormal agers

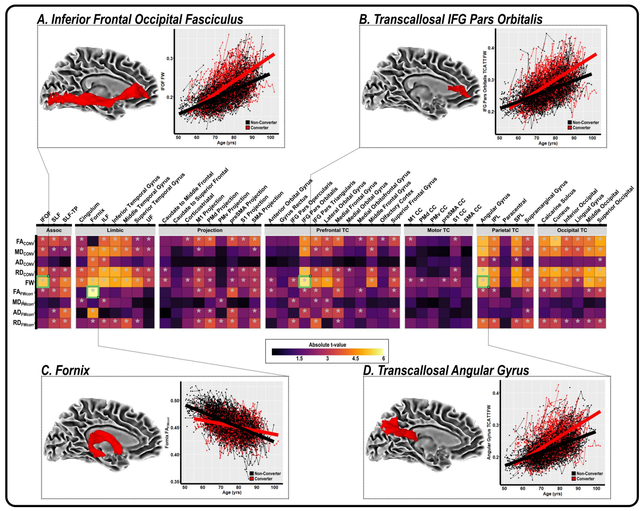

Figure 2. A heatmap displays the strength of interactions across all white matter features. Strong effects were seen in the inferior frontal occipital fasciculus (**A**), transcallosal inferior frontal gyrus pars orbitalis (**B**), fornix (**C**), and transcallosal angular gyrus (**D**).

Finding 3: Considering conversion along the AD continuum significant increases predicted variance in aging models

Table 2. Marginal R^2 in Age and AD Aware Models

Model	Conventional				FW-Corrected				
	FA_{CONV}	MD_{CONV}	AD_{CONV}	RD_{CONV}	FW	FA_{FWcorr}	MD_{FWcorr}	AD_{FWcorr}	RD_{FWcorr}
Age	14.57 (6.37)	27.81 (5.94)	24.74 (9.05)	25.50 (6.92)	27.59 (4.44)	9.10 (4.65)	4.29 (5.45)	7.02 (5.04)	7.12 (4.89)
AD Aware	17.67 (6.99)	31.54 (5.96)	26.99 (9.68)	29.55 (7.25)	32.03 (4.55)	10.79 (5.07)	4.95 (6.01)	8.32 (5.21)	8.43 (5.13)
Difference	3.10 (1.61)	3.73 (1.83)	2.25 (1.52)	4.05 (1.96)	4.44 (1.91)	1.70 (1.11)	0.66 (0.64)	1.30 (0.85)	1.31 (0.93)

Finding 4: FW within limbic tracts is most vulnerable to abnormal aging

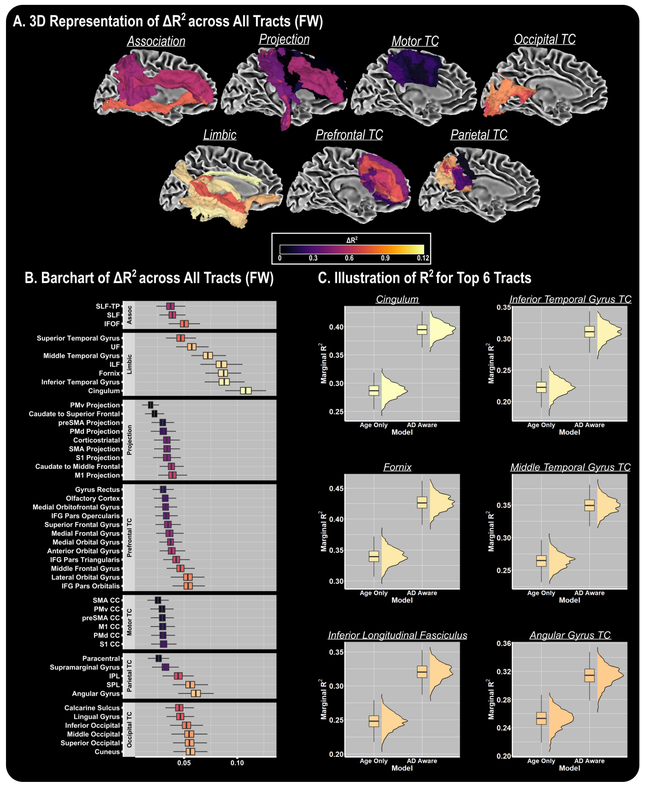

Figure 3. Tracts are highlighted based on their FW ΔR^2 (**A**) and bar charts for ΔR^2 are shown (**B**). Marginal R^2 for the age and AD aware models are shown for the top 6 tracts.

Conclusions

- Significant interactions on white matter microstructure exist between normal and abnormal agers, with predominant interactions in several association, limbic, prefrontal transcallosal and parietal transcallosal tracts.
- A bootstrapped analysis of "Age" and "AD Aware" models found that there were significant differences when considering AD diagnosis and conversion along the AD continuum. This analysis specifically demonstrated that:
  1) The FW metric is most vulnerable to abnormal aging.
  2) FW in the limbic tracts are most vulnerable to abnormal aging, whereas motor transcallosal tracts shown little effect from abnormal aging.

The findings of this study suggest that FW correction be conducted when using single-shell diffusion MRI acquisition studies to evaluate abnormal aging. Future studies should pair this technique with genomic data to further understand the biological mechanisms driving white matter neurodegeneration in aging.

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